

DRÆM: A Discriminatively Trained Reconstruction Embedding for Surface Anomaly Detection

Scientific Report

Seminar Paper

in the context of the seminar “Advanced Machine Learning for Anomaly Detection”

at Friedrich-Alexander-Universität Erlangen-Nürnberg
at the Department Artificial Intelligence in Biomedical Engineering (AIBE)
Image Data Exploration and Analysis (IDEA) Lab

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My Extensions and Experiments

My work extends the code provided by the DRÆM paper (Zavrtanik et al., 2021), primarily through additional anomaly generation methods¹. To run more experiments and also use the lower-end GPUs of the HPC, I implemented micro-batch training and used automatic mixed precision (AMP) for the forward pass, while using full precision for loss calculation and evaluation to ensure numerical stability. In addition, I fixed a bug where the last incomplete batch of each epoch was weighted incorrectly during training, which was causing training instability. I also switched to a cosine annealing learning rate scheduler to ensure comparability between different methods without the need for additional hyperparameter searches. Multiple test runs were performed after each adjustment to rule out any negative impact on the final performance. In fact, my reproduced AUROC metrics (image- and pixel-level) on the combined MVTec dataset are within 0.1 percentage points of the values reported in the paper, while the pixel AP value improved from 68.4% to 69.7%, likely due to more stable training.

I observed that the anomaly blending method in the paper always produced anomalies with sharp borders, which are relatively easy to detect. By varying the anomaly strength throughout the anomaly region, more realistic anomalies can be generated, which should lead to a more robust model. For all blending methods, I kept the anomaly mask generation the same and varied only the strength depending on the blending method. The blending methods are: uniform β blending (as in the original paper, without spatial variation in anomaly strength), Perlin- β blending (where Perlin noise determines the anomaly opacity), texture- β blending (where the luminance of a randomly sampled texture determines the anomaly opacity), blurred β blending (where the borders of the anomaly mask are blurred), and channel-wise Poisson image interpolation, suggested by my supervisor, Johanna Müller, where the anomaly strength is guided by the source image gradient (see Tan et al. (2021)). The Poisson image interpolation, as the only content-sensitive blending method, yields on average the best results. However, on some object categories, the other methods outperform Poisson interpolation. The best general performance was achieved by combining the Poisson interpolation and Perlin- β blending methods, randomly choosing one for each generated anomaly.

Method	Image AUROC $\uparrow$	Pixel AUROC $\uparrow$	Pixel AP $\uparrow$
DRÆM-Uniform	98.0%	97.3%	69.7%
DRÆM-Blurred	96.4%	96.0%	65.8%
DRÆM-Texture	98.0%	97.2%	69.1%
DRÆM-Perlin	98.2%	97.3%	69.5%
DRÆM-Poisson	99.1%	97.8%	71.6%
DRÆM-Poisson+Perlin	99.2%	98.0%	73.6%

¹ My full code is available at https://github.com/christopher-rossbach/DRÆM_seminar_AD.

References

- Tan, J., Hou, B., Day, T., Simpson, J., Rueckert, D., & Kainz, B. (2021). Detecting Outliers with Poisson Image Interpolation. In M. de Bruijne, P. C. Cattin, S. Cotin, N. Padoy, S. Speidel, Y. Zheng, & C. Essert (Eds.), *Medical Image Computing and Computer Assisted Intervention – MICCAI 2021* (pp. 581–591). Springer International Publishing. https://doi.org/10.1007/978-3-030-87240-3_56
- Zavrtanik, V., Kristan, M., & Skocaj, D. (2021). Draem - a discriminatively trained reconstruction embedding for surface anomaly detection. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 8330–8339.

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